

SkyGuard AI • Executive System Architecture & Feature Dossier

Autonomous Meteorological Quality Control, Anomaly Discrimination & Tactical Evacuation Platform

Initiative: Smart India Hackathon 2026

Problem ID: 26073 (Disaster Management)

Team: AI Avengers

Authority: MoES / IMD AWS Grid

1. Executive Summary & Core Problem Statement

Across India's national grid of **Automatic Weather Stations (AWS)** operated by the **India Meteorological Department (IMD)**, State Disaster Management Authorities (SDMAs), and agricultural departments, field sensors endure harsh environmental stressors: monsoonal flash floods, coastal salt incrustation, extreme summer heatwaves, and physical debris.

This environment introduces a frequent and critical operational failure mode: **The Sensor Fault vs. Genuine Extreme Weather Dilemma**. When an isolated surface sensor suddenly transmits an extreme anomaly (e.g., a 55°C heat spike or a -15 hPa pressure plummet), traditional rule-based threshold systems fail to determine if:

- **Scenario A (Hardware Malfunction):** An open circuit on a PT100 RTD temperature probe, a stuck ADC register, a desiccated capacitive polymer hygrometer, or a spider web in a tipping-bucket rain gauge.
- **Scenario B (Genuine Extreme Disaster):** A real mesoscale convective microburst, squall line, cloudburst, cyclonic landfall, or catastrophic flash flood.

The SkyGuard AI Solution: SkyGuard AI solves this challenge through a multi-tier physics-grounded AI engine. It continuously evaluates thermodynamic consistency (Clausius-Clapeyron relations), spatial peer consensus (K-Nearest Neighbors buddy networks), and temporal rate-of-change limits. It automatically quarantines corrupted telemetry from Numerical Weather Prediction (NWP) model feeds, substitutes mathematically imputed values via Inverse-Distance Weighting (IDW), generates WMO-No. 8 Standard Operating Procedures (SOPs), and dispatches predictive disaster evacuations.

2. High-Level Technical Architecture

Subsystem Layer	Technologies & Protocols	Primary Operational Responsibilities
Frontend Mission Control	HTML5, Modern CSS Variables, Modular Vanilla JS (ES Modules), Chart.js 4.4, Leaflet GIS 1.9	Real-time dashboard, high-definition satellite/topo map, sensor health radar, batch CSV audit workbench, tactical evacuation routing UI, interactive copilot drawer.
Secure Server Proxy	Python HTTP Server (<code>server.py</code>), Vercel Serverless Function (<code>api/chat.js</code>)	Keeps private API keys safely on the server, encloses filesystem in a strict directory jail, blocks path traversal and dotfiles, enforces 20 req/min sliding-window IP rate limiting.
Authentication & Access Control	HMAC-SHA256 Signed Session Tokens, Bearer Token Validation	Authorizes IMD/SDMA duty officers, secures privileged LLM intelligence pipelines, prevents unauthorized quota drain.
AI Intelligence Engine	Google Gemini 2.5 Flash API + SkyGuard Offline Physics Diagnostic Engine	Context-aware conversational diagnostics, sensor failure root-cause analysis, disaster management SOPs, executive briefings.

3. Comprehensive Breakdown of System Modules

Module 1: Global Command Bar & Operator Access Control

- **Live Ingestion Stream Controller:** Real-time streaming status with single-click pause/resume capability to freeze live frames for forensic review.
- **AWS Node Switcher:** Instant switching between high-risk national AWS nodes (Bangalore IMD, Sakleshpur Western Ghats, Chennai Coastal, Mumbai Santacruz, Leh High-Altitude, Cherrapunji).
- **Stream Data Provider:** Toggle between live **Open-Meteo REST API feeds** and the high-fidelity **Department Telemetry Simulator**.
- **Network Operational Trust Gauge:** Aggregate index reflecting instantaneous data credibility and thermodynamic validity across all active sensors.
- **Operator Access Control (HMAC Auth):** Session authentication modal for authorized personnel with timing-safe HMAC validation.

- **Emergency Global Alert Ribbon:** Automatically appears at the top of the interface whenever high-risk natural hazard precursors are detected.

Module 2: Network Command Center (Dashboard View)

- **Real-Time Atmospheric KPIs:** High-precision readouts for Dry-Bulb Temperature (°C), Relative Humidity (%), Barometric Pressure (hPa), Wind Speed & Gusts (km/h), Rainfall Accumulation (mm), and Dew Point Depression.
- **5-Tier AI Quality Control Verdicts:**
 - **Flag G (NORMAL):** Nominal operation within standard physical tolerance envelopes.
 - **Flag F (FAULT):** Instrumentation fault detected and quarantined from downstream NWP models.
 - **Flag S (EVENT):** Genuine extreme weather event corroborated by regional spatial buddy stations.
 - **Flag W (WATCH):** Sensor degradation or progressive calibration drift identified.
- **Virtual Sensor Imputation Widget:** Reconstructs real-time substitute values using Inverse-Distance Spatial Weighting (IDW) from peer stations when a reading is quarantined.
- **Live Alerts Feed:** Chronological audit table detailing time, station ID, QC flag, severity level, and plain-language root-cause reasoning.
- **Thermodynamic Consistency Chart:** Interactive dual-axis curve plotting Temperature vs. Dew Point to enforce Clausius-Clapeyron limits.

Module 3: Spatial AWS Network Map View

- **High-Definition Multi-Layer Basemaps:** Switch between **Esri World High-Res Satellite Imagery**, **OpenTopoMap Topographic Contours**, **CartoDB Dark Matter**, and **Standard Street Maps**.
- **Dynamic Sensor Health Markers:** Real-time color-coded station markers pulsing according to AI health state.
- **Spatial Peer Consensus Interconnects:** Visual geometric vectors illustrating active K-Nearest Neighbor (KNN) correlation links between target stations and regional peers.
- **Interactive Station Dossiers:** Interactive popups displaying coordinates, altitude, sensor models, live telemetry, and quick-action links.

Module 4: Telemetry & Multi-Parameter Analytics

- **Multi-Parameter Time-Series Chart:** 24-hour synoptic curves synchronizing Temperature, Relative Humidity, Pressure, Wind, and Rain.
- **Raw vs. Imputed Comparative Analysis:** Direct visual overlay contrasting corrupted raw telemetry against the AI-reconstructed Virtual Sensor curve.
- **Spatial Buddy Comparison:** Evaluates target station departures against regional consensus curves.

- **Transducer Health Radar:** 5-axis radar chart showing individual health scores across the RTD probe, capacitive hygrometer, barometer, anemometer, and ESP32 telemetry bus.

Module 5: Simulation & Department Ingestion Workbench

- **Interactive Hardware Fault & Extreme Weather Injection:**
 - *55°C Electrical Spike:* Simulates open RTD circuit / voltage transient.
 - *Frozen Sensor Signal:* Simulates stuck ADC register lock on a capacitive hygrometer.
 - *-14 hPa Barometric Drop:* Simulates pressure port dust blockage without weather correlation.
 - *ESP32 Uplink Dropout:* Simulates telemetry comms timeout and dropped packets.
 - *Calibration Drift:* Simulates gradual thermal sensor degradation over time.
 - *Severe Regional Thunderstorm:* Simulates real mesoscale storm across multiple peer stations.
- **Department Batch CSV/JSON Ingestion Engine:** Ingests thousands of department historical AWS telemetry records with a built-in stress-test data generator.
- **Audited CSV Export:** Exports fully audited CSV files with imputed values and WMO QC flags, protected against CSV Formula Injection (CWE-1236).

Module 6: Sensor Health Matrix & WMO-No. 8 Standard Operating Procedures (SOP)

- **Component-Level Diagnostics:** Real-time health percentages, calibration dates, and model numbers for PT100 RTD probes, capacitive humidity chips, Vaisala barometers, and ultrasonic anemometers.
- **Dynamic WMO-No. 8 SOP Generator:** Automatically prescribes certified engineering maintenance procedures:
 - 2-point salt chamber calibration (LiCl 11.3% RH and NaCl 75.3% RH) for hygrometer drift.
 - Radiation shield cleaning and aspiration filter checks for dust/salt blockages.
 - Lead-wire resistance balancing and excitation voltage verification for RTD sensors.

Module 7: Predictive Disaster Ops & Tactical Evacuation Routing

Predictive Early Warning Horizon: Detects severe natural hazard precursors **6 to 72 hours ahead of catastrophic impact.**

- **Supported Disaster Hazards:**
 - **Landslides:** Tracks antecedent 72-hour rainfall saturation, slope steepness, and soil moisture thresholds (e.g., Western Ghats / Sakleshpur corridor).

- **Flash Floods:** Detects precipitation rate spikes exceeding hydrological drainage capacity.
- **Cyclonic Squalls:** Detects barometric plunge rates exceeding -3 hPa/hr coupled with severe squall gusts.
- **Severe Heatwaves:** Flags prolonged temperatures exceeding +4.5°C above seasonal normal.
- **Tactical Evacuation Corridor Engine:** Computes the shortest and safest evacuation route from danger zones to certified relief shelters, avoiding floodplains, unstable slopes, and surge zones.
- **Multi-Agency Emergency Broadcast Dispatcher:** Simulates one-click emergency cell broadcast transmission across SDMA, NDRF, District Police, and public citizen SMS alerts.

Module 8: Official Forensic Quality Control Audit Dossier

- **Government Audit Dossier:** Generates an official, publication-ready forensic quality certification report for IMD / MoES records.
- **Mathematical Proof Box:** Formal proof validating Clausius-Clapeyron thermodynamic consistency, spatial buddy consensus, and IDW imputation trails.
- **Executive Print Preview:** Built-in print CSS formatted for official sign-off and archiving.

Module 9: SkyGuard AI Weather Copilot

- **Interactive Floating Robot:** Animated desktop robot companion that expands into a conversational AI drawer.
- **Google Gemini 2.5 Flash Engine:** Proxies multi-turn queries through the secure backend to answer meteorological, hardware, electronic, mathematical, and disaster response questions.
- **Zero-Key Offline AI Fallback:** Automatically falls back to an internal offline rule-based diagnostic engine if the backend or internet connection is unavailable.

Module 10: Production Security Hardening & Zero-Leakage Architecture

- **Operator Authentication Layer:** HMAC-SHA256 signed session tokens with secure verification protecting privileged endpoints.
- **Sliding-Window IP Rate Limiter:** Enforces a 20 request / 60-second limit on `/api/chat` to prevent quota exhaustion and DoS attacks.
- **Directory Traversal Defense:** Prevents URL-encoded traversal (`%2e%2e`, `%2eenv`) and enforces root directory path jailing.
- **Zero Secret Leakage:** Complete redaction of API keys from frontend and configuration health checks (`/api/config`).
- **XSS & CSV Formula Sanitization:** Universal HTML escaping on all dynamic table elements and single-quote escaping on formula trigger characters (`=`, `+`, `-`, `@`).

